



# DIPESH PAUL

Senior Machine Learning Engineer | GenAI & LLM Specialist

Indore, MP, 452016 | +91 896 277 5625

dipeshpal17@gmail.com | <https://dipeshpal.in>

## PROFESSIONAL SUMMARY

Senior Machine Learning Engineer (7 years) specializing in GenAI, LLMs, multi-agent systems, RAG, and model finetuning. Experienced with OpenAI, DeepSeek, Llama, Mistral, QLoRA, PEFT, distributed training, and vector databases (Weaviate, Pinecone, Qdrant). Built and deployed production-grade ML, DL, and LLM systems using FastAPI, Docker, and cloud services, delivering scalable, low-latency AI solutions in enterprise environments.

## SKILLS

- |                                             |                |                        |                          |
|---------------------------------------------|----------------|------------------------|--------------------------|
| • GenAI                                     | • Tensorflow   | • PydanticAI           | • Jira                   |
| • Machine Learning                          | • Keras        | • Langchain            | • Git                    |
| • Deep Learning                             | • Pytorch      | • NLP                  | • Agent Frameworks       |
| • LLM OpenAI, DeepSeek, Llama, Mistral etc. | • Scikit Learn | • SFT, RLHF, Peft etc. | • Kafka                  |
| • LLM Finetuning                            | • Transformers | • Docker               | • Team Lead              |
| • OpenAI Agent SDK                          | • Ollama       | • MCP & Tools          | • Client Handling        |
| • LlamaIndex                                | • OpenCV       | • Function Calling     | • CrewAI                 |
| • Pinecone, Qdrant, Weaviate                | • Spacy        | • Embeddings           | • N8N                    |
|                                             | • RAG          | • Prompt Engineering   | • Architecture Designing |

## EXPERIENCE

### Systango, Indore (Senior Machine Learning Engineer)

Dec 2023 - Present

- Designed and implemented distributed finetuning workflows using QLoRA, reducing training time by ~30% and cutting training costs by ~20%.
- Built multi-agent orchestration using OpenAI Agent SDK + PydanticAI for real-time task routing, improving routing accuracy and reducing manual triage effort by 40%.
- Collaborated with cross-functional teams (product, infra) to productionize LLM features with FastAPI and Docker, ensuring low-latency, high-availability deployments.

### Zdaly, Indore (Senior Machine Learning Engineer)

Feb 2023 - July 2023

- Built an LLM-powered product catalog analyzer that improved lookup accuracy by 45%.
- Designed a vector-based similarity search pipeline using Weaviate for real-time customer recommendations.
- Migrated legacy NLP models to transformer-based architectures, reducing latency by 50%.

### GsLab, Pune (Machine Learning Engineer)

Jan 2022 - Jan 2023

- Developed a ticket classification model using ULMFit + FastAI that improved routing accuracy by 38%.
- Automated price prediction system using TensorFlow & Scrapy, processing 100k+ product datapoints.
- Optimized inference pipeline with Dockerized microservices, cutting server cost by 25%.

### Calsoft, Pune (Machine Learning Engineer)

Aug 2019 - Jan 2022

- Built an avatar generator using GANs (DCGAN, StyleGAN) trained on 143k+ images.
- Implemented virtualization automation PaaS platform using TensorFlow. Reduced VM provisioning time by 27%.
- Developed ML pipelines for anomaly detection & forecasting using Scikit-Learn + Kafka.

### Calsoft, Pune (Machine Learning Engineer Intern)

Jan 2019 - Jul 2019

- Created internal POCs in image classification, OCR, and NLP.
- Built early prototypes of ticket analyzer using traditional NLP + Python.
- Contributed to optimizing CNN models, improving accuracy by 12%.

## KEY PROJECTS

### Multi-Agent Procurement Platform

- Built a Multi-Agent platform to streamline procurement, featuring LLM-powered chatbot portals for support, order analysis, and data visualization. It collects user requirements and routes requests to suitable vendors.
- OpenAI's GPT-4o, GPT-4o-mini, Weaviate DB, Postgres, FastAPI, OpenAI Agent SDK, PydanticAI

*Python, OpenAI Agent SDK, Weaviate, Postgres, FastAPI, PydanticAI, Project Lead, Architecture Design*

### Automated Customer Support LLM Powered

- Engineered an AI-powered customer support system that intelligently connects customers to specialized bots and agents, providing 24/7 query resolution and support.

*OpenAI's GPT-4.1, FastAPI, Weaviate vector DB, Architecture Design*

### LLM Powered Productivity Bot

- Built a comprehensive platform allowing users to connect and sync data from applications like Gmail, Calendar, and Outlook. The system delivers actionable insights, recommendations, and cross-app search capabilities using embedding models and similarity searches.

*OpenAI's GPT-4o, FastAPI, Weaviate vector DB, AWS Bedrock, OpenAI API, RAG*

### Product Chat Agent

- Created an advanced chatbot that leverages large language models to provide detailed product information and streamline purchase processes based on natural language user prompts.

*Python, Langchain, OpenAI, FastAPI, Prompt Engineering*

---

## ADDITIONAL PROJECTS

### Price Analysis with Scraping

- Built a comprehensive price analysis system that scrapes data from global E-commerce and OEM websites to inform pricing strategies for computing devices. Implemented machine learning models for accurate price prediction.

*Python, FastAPI, AI/ML, Tensorflow, Jenkins, Docker, Scrapy, Transformers*

### Automation of Virtualization

- Developed a PaaS tool that automates virtualization solutions, enabling customers to create virtual machines and deploy applications effortlessly. The platform includes robust monitoring, troubleshooting capabilities, and AI-based price prediction.

*Python, FastAPI, AI/ML, Tensorflow, Jenkins, Docker, Transformers*

### Ticket Analyzer

- Automated ticketing processes using advanced machine learning techniques. Utilized NLP and Wikipedia-trained ULMfit Fastai models to analyze tickets and intelligently determine appropriate assignees.

*Python, Kafka, Docker, FastAPI, AI/ML, Fastai*

### Avatar Generator

- Built an avatar generation system for portfolio faces, trained on a dataset of approximately 143,000 human and cartoon images. Implemented various neural network architectures, including Deep Convolutional GANs (Style GAN), CNN, and UNet.

*Python, AI/ML, Tensorflow, OpenCV, DZone*

---

## AWARDS

- Rising Star, 2020, Calsoft Pvt. Ltd.
- Outstanding Innovator, 2022, Calsoft Pvt. Ltd.
- Praise, 2024, Systango

## CERTIFICATIONS

- **Machine Learning:** Stanford University (96.47%)
  - **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning:** Coursera (99.28%)
  - **Supervised Machine Learning (Regression and Classification):** DeepLearning.AI, Stanford University (100%)
  - **Python Data Structures:** University of Michigan (98.40%)
- 

## EDUCATION

### RGPV University

Bachelor of Engineering in Computer Science, 2019

Acropolis Institute of Tech. & Research, Indore

**CGPA:** 7.5

### M.P. Board

Higher School, 2015

**Score:** 75%

### M.P. Board

Secondary School, 2013

**Score:** 80%

---

## OPEN-SOURCE & PUBLICATIONS

### Chatformers

- chatformers: Chatformers is a Python package that simplifies chatbot development by automatically managing chat history using local vector databases like Chroma DB.
- Pypi URL: [pypi.org/project/chatformers](https://pypi.org/project/chatformers)

### JarvisAI

- JarvisAI: JarvisAI is Python library to build your own AI virtual assistant with natural language processing.
- Pypi URL: [pypi.org/project/JarvisAI](https://pypi.org/project/JarvisAI)

### Python is All You Need – ebook covering Python from basics to advanced

- Ebook URL: [techport.myinstamojo.com/product/python-is-all-you-need](https://techport.myinstamojo.com/product/python-is-all-you-need)
- 

## GITHUB STATS

- **Total Repository:** 104
  - **Stars Count:** 639
  - **Commits Count:** 2.1K
  - **Followers:** 97
  - **Stats Checker:** <https://gh-readme-profile.vercel.app/api?username=Dipeshpal>
- 

## LINKS

- **LinkedIn:** <https://www.linkedin.com/in/dipesh-paul>
- **GitHub:** <https://github.com/Dipeshpal>
- **Portfolio:** [www.dipeshpal.in](http://www.dipeshpal.in)